



INTRODUCTION TO DATA COMMUNICATION AND COMPUTER NETWORK (IDCCN) **[CSIT369]**

MODULE 1

INTRODUCTION TO SIGNALS

(UG - III Sem)

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INTRODUCTION TO DATA COMMUNICATION

- Communication is defined as a process in which more than one computer transfers information, instructions to each other and for sharing resources.
- Or in other words, communication is a process or act in which we can send or receive data.
- A network of computers is defined as an interconnected collection of autonomous computers.
- Autonomous means no computer can start, stop or control another computer.

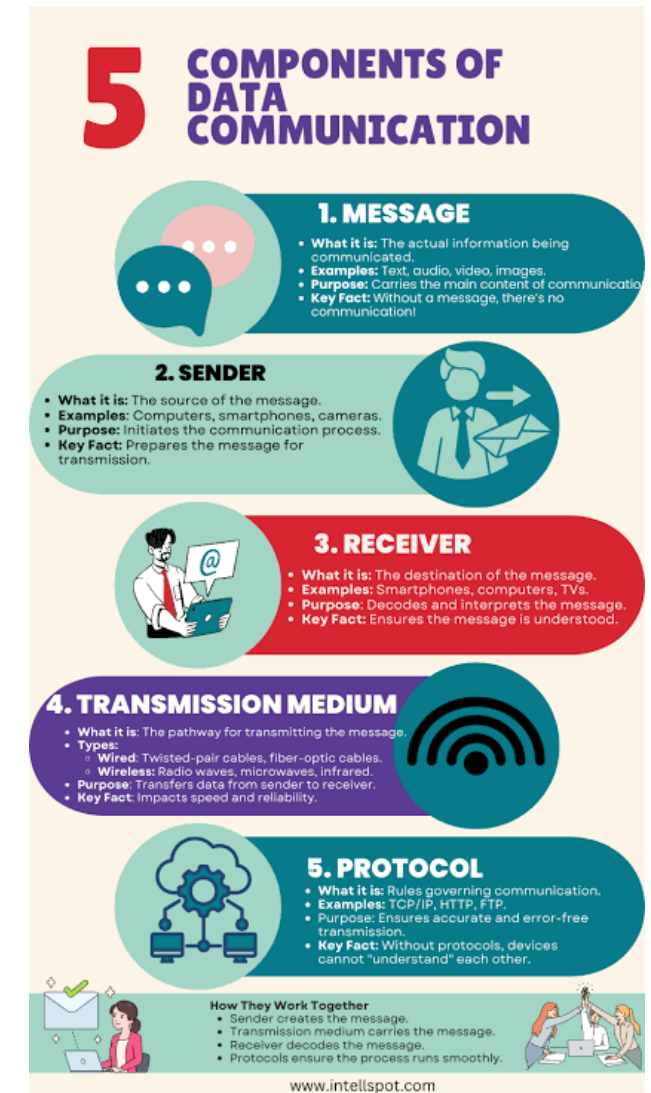


CHARACTERISTICS OF DATA COMMUNICATION

- **Delivery** – Data must reach the correct destination. (MAC address, IP address)
- **Accuracy** – Data must be delivered without errors. Altered data is unusable.
- **Timeliness** – Data must arrive on time. In the case of video and audio, late delivery is useless. In the case of video and audio, late delivery is useless.
- **Jitter** – Variation in packet arrival time should be minimal. It is the delay in the delivery of audio or video data packets. an uneven quality in the video is the result.

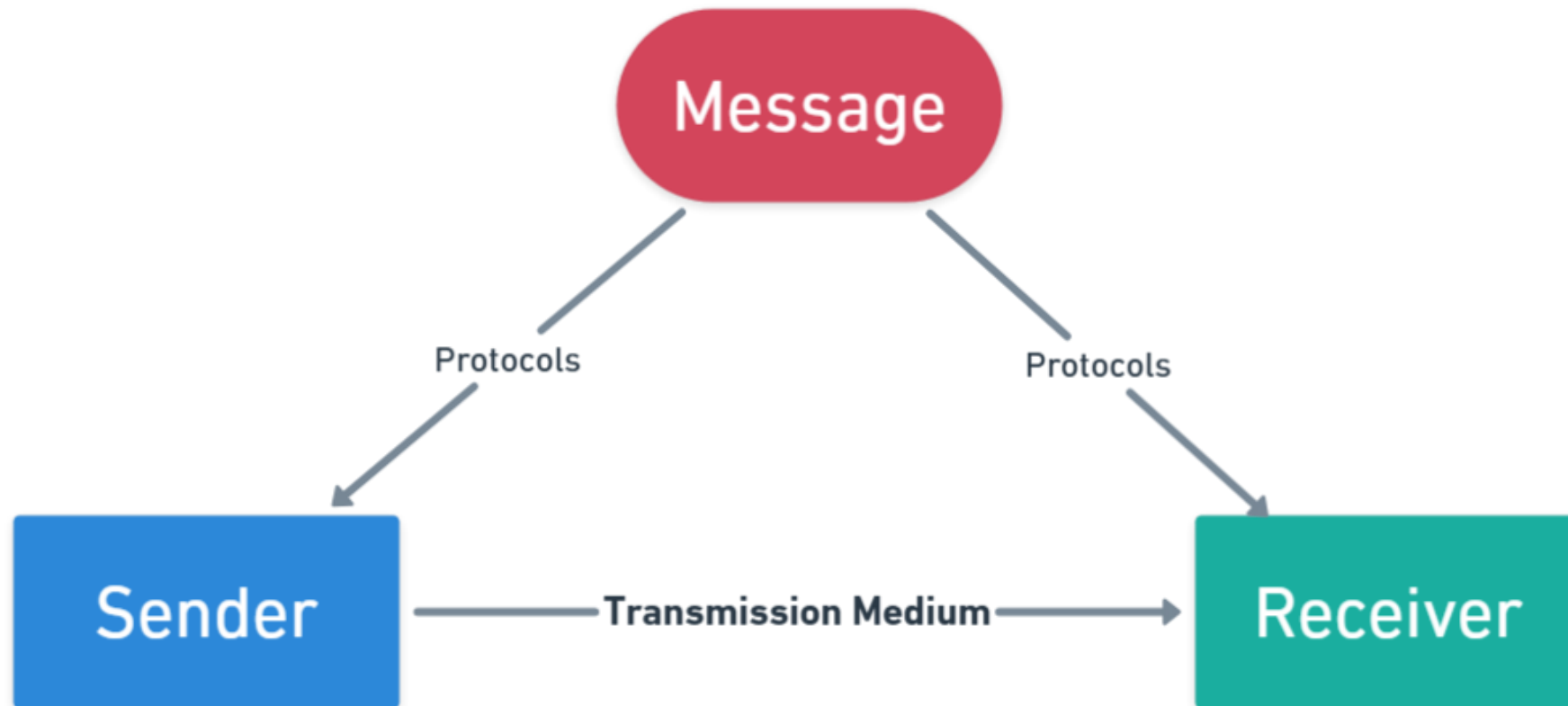
COMPONENTS OF DATA COMMUNICATION

1. **Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.
2. **Sender:** It is simply a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.
3. **Receiver:** It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.
4. **Transmission Medium / Communication Channels:** Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless Media.
5. **Protocols:** Set of rules governing communication. Example: TCP/IP





COMPONENTS OF DATA COMMUNICATION





DATA REPRESENTATION

Computers represent all types of information using binary digits (**0** and **1**). Different forms of data are converted into binary format before they can be processed, stored, or transmitted.

Why Data Representation?

- Enables computers to process different types of data.
 - Provides a standard format for storage and communication.
 - Forms the foundation of digital systems and computer networks.
-
- **Text**
 - **ASCII (American Standard Code for Information Interchange).**
 - Example: A = 65, B = 66, C = 67.
 - **Number**
 - **Decimal System:** Uses digits 0–9 (Example: 25).
 - **Binary System:** Uses only 0 and 1 (Example: $25 = 11001_2$).



DATA REPRESENTATION

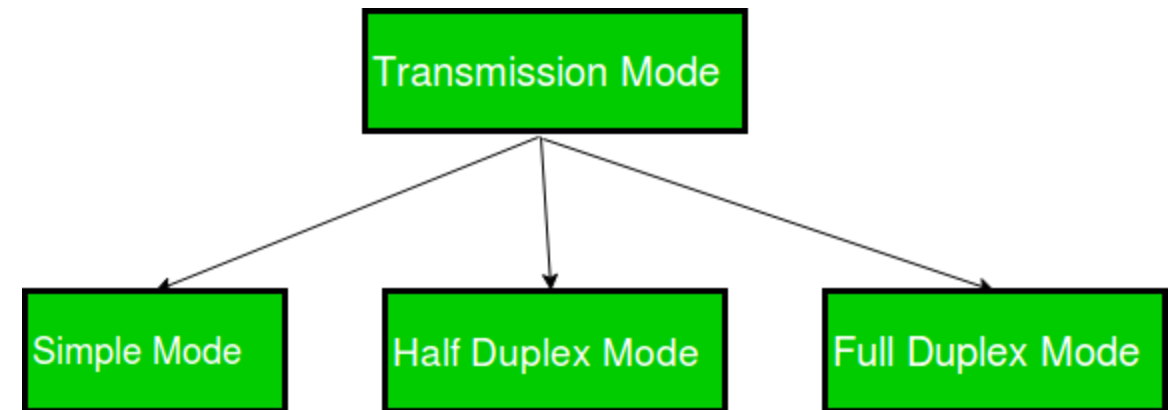
- **Images**
 - Images are represented using pixels.
 - Example: Resolution: 1920×1080 , Color Depth: 24-bit
 - Higher resolution $\rightarrow$ Better quality
- **Audio**
 - Sampled and digitized
 - Example: MP3, WAV
- **Video**
 - Sequence of images (frames)
 - Example: 30 frames/sec
 - Formats: MP4 AVI MOV

Key Point: Regardless of whether the data is text, numbers, images, audio, or video, it is ultimately represented in binary form inside a computer.



DATA FLOW

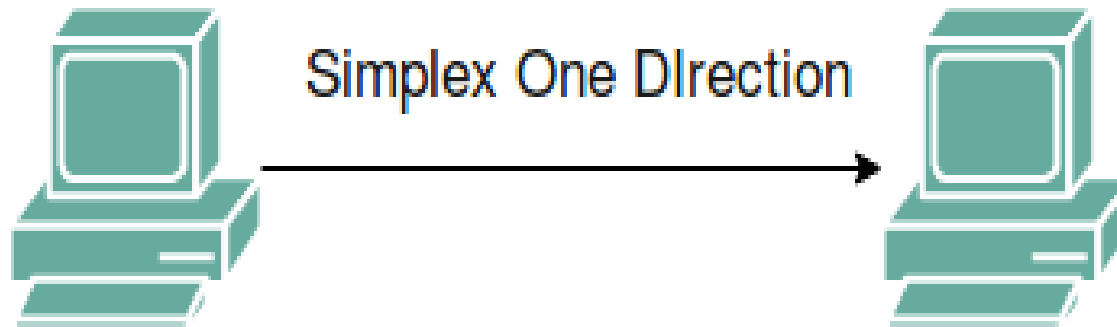
- Data flow determines the direction of communication between devices.
- It is also known as a communication mode.
- Buses and networks are designed to allow communication to occur between individual devices that are interconnected.
- Types:
 1. Simplex
 2. Half Duplex
 3. Full Duplex





SIMPLEX

- In Simplex mode, the communication is unidirectional, as on a one-way street.
- Only one of the two devices on a link can transmit, the other can only receive.
- The simplex mode can use the entire capacity of the channel to send data in one direction.
- Example: Keyboard and traditional monitors.
- The keyboard can only introduce input, the monitor can only provide the output.



SIMPLEX

Advantages

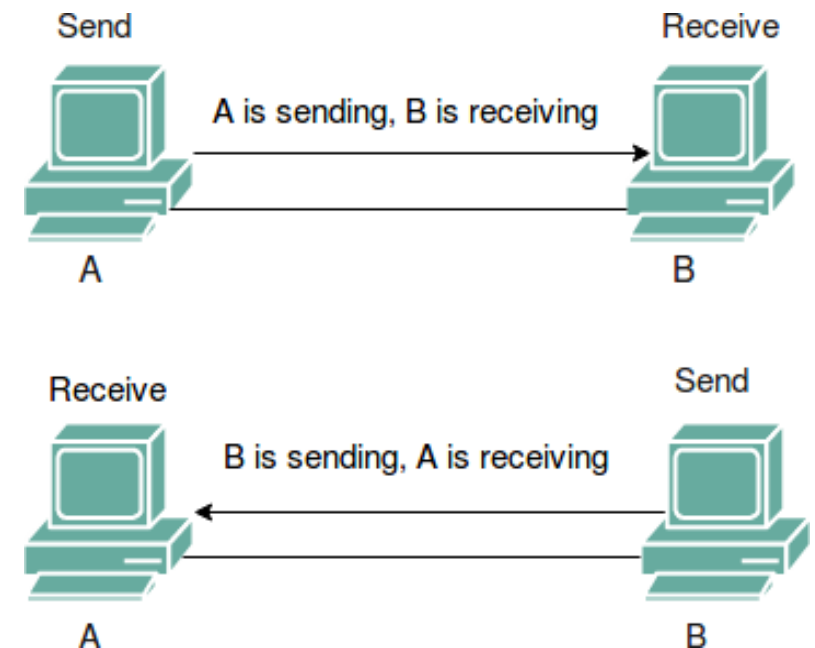
- Simple and reliable communication method.
- Cost-effective as only one communication channel is required.
- No coordination is needed between sender and receiver.
- Suitable for applications where feedback is not required.
- Commonly used in broadcasting and surveillance systems.

Disadvantages

- Supports communication in only one direction.
- Receiver cannot send feedback to the sender.
- No mechanism to verify correct data reception.
- Not suitable for interactive or bidirectional communication.
- Limited flexibility compared to half-duplex and full-duplex modes.

HALF DUPLEX

- In half-duplex mode, each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time.
- The entire capacity of the channel can be utilized for each direction.
- Example: Walkie-talkie in which message is sent one at a time and messages are sent in both directions.





HALF DUPLEX

Advantages

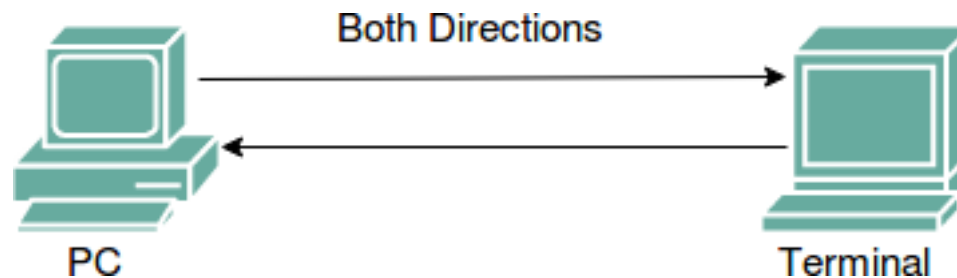
- Supports bidirectional communication, allowing devices to both send and receive data.
- More efficient than simplex communication, as the same channel is used for both transmission and reception.
- More cost-effective than full-duplex communication because only one communication channel is required.

Disadvantages

- Devices cannot transmit and receive data simultaneously.
- Communication delays may occur due to alternating transmission and reception.
- Requires coordination between communicating devices, which can increase complexity.
- Less efficient and reliable than full-duplex communication for real-time applications.

FULL DUPLEX

- In full-duplex mode, both stations can transmit and receive simultaneously.
- In full duplex mode, signals going in one direction share the capacity of the link with signals going in another direction, this sharing can occur in two ways:
 - Either the link must contain two physically separate transmission paths, one for sending and the other for receiving.
 - Or the capacity is divided between signals traveling in both directions.





FULL DUPLEX

Advantages

- Supports simultaneous bidirectional communication, allowing devices to send and receive data at the same time.
- Ideal for real-time applications such as telephone calls, video conferencing, and online gaming.
- Provides the highest communication efficiency, as transmission and reception occur simultaneously.
- Offers greater reliability and faster communication with minimal delays.

Disadvantages

- More expensive than simplex and half-duplex communication because it requires two communication channels or equivalent resources.
- More complex to implement and manage due to the need for separate transmission and reception paths.
- Requires higher bandwidth and network resources.
- May not be necessary for applications where simultaneous two-way communication is not required.

SIGNALS

A signal is a physical quantity that carries information from one point to another. In communication systems, signals are used to transmit data through various media such as wires, optical fibres, and wireless channels.

Importance of Signals

- Enable communication between devices.
- Carry information such as text, voice, images, and video.
- Form the foundation of all modern communication systems.

Examples

- Human speech transmitted as sound waves.
- Mobile communication using radio waves.

Key Point: Based on the nature of the transmitted waveform, signals are broadly classified into **Analog Signals** and **Digital Signals**.

ANALOG SIGNALS

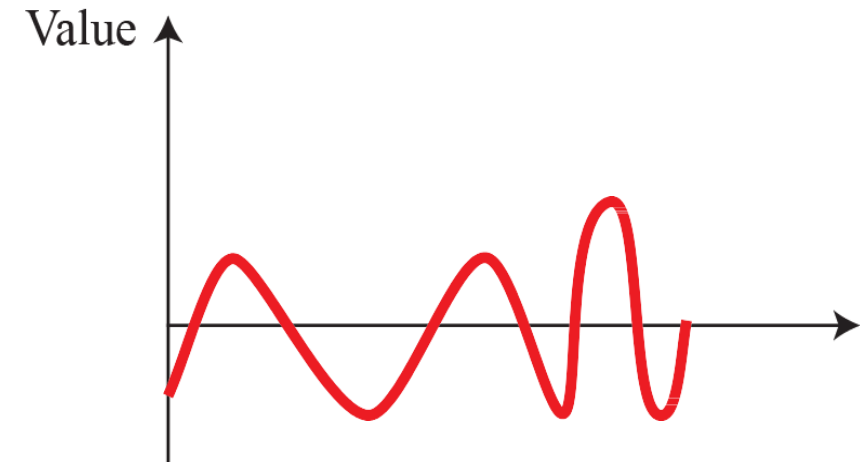
An analog signal is a continuous signal that varies smoothly with time and can take any value within a given range. Most naturally occurring signals are analog in nature.

Characteristics

- Continuous in time and amplitude.
- Can have infinitely many values.
- Represent natural phenomena.
- Usually shown as smooth waveforms.

Examples

- Human voice.
- Temperature variations.
- Traditional radio broadcasts.



a. Analog signal

Key Point: Analog signals vary continuously and represent real-world information in its natural form.



DIGITAL SIGNALS

A digital signal is a discrete signal that represents information using a finite number of values, typically 0 and 1. Digital signals form the basis of modern communication and computer systems.

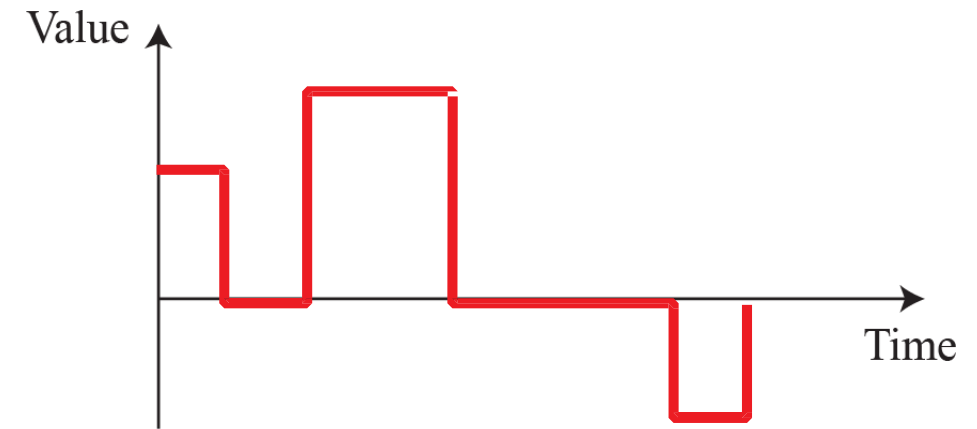
Characteristics

- Discrete in time and amplitude.
- Consists of binary values (0s and 1s).
- Less affected by noise and distortion.
- Easy to store, process, and transmit.

Examples

- Computer data.
- Mobile phone communication.
- Digital audio and video.

Key Point: Digital signals represent information using discrete levels, making them more reliable and efficient for modern communication systems.



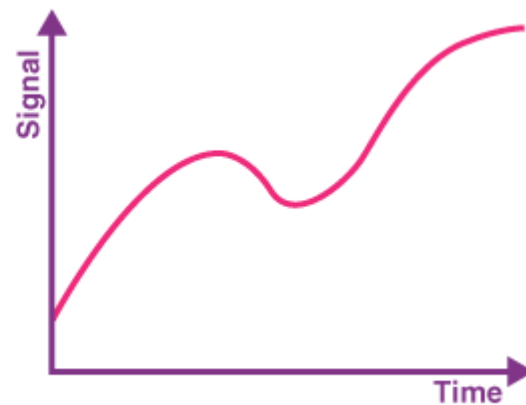
b. Digital signal



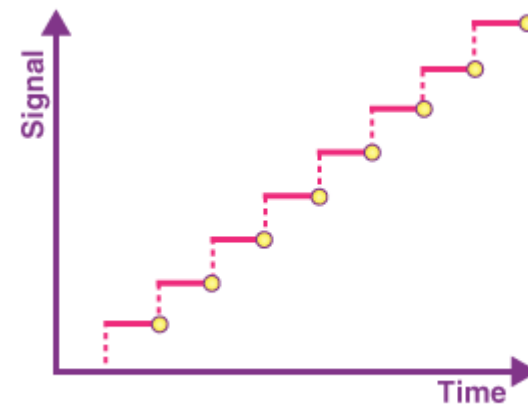
ANALOG SIGNAL V/S DIGITAL SIGNALS

Data can be analog or digital. The term analog data refers to information that is continuous; digital data refers to information that has discrete states. For example, an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous. On the other hand, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.

Like the data they represent, signals can be either analog or digital. An analog signal has infinitely many levels of intensity over a period of time. As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path. A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.



Analog signal



Digital signal



ANALOG SIGNAL V/S DIGITAL SIGNALS

Difference between Analog and Digital Signal	
Analog Signals	Digital Signals
Continuous signals	Discrete signals
Represented by sine waves	Represented by square waves
Human voice, natural sound, analog electronic devices are a few examples	Computers, optical drives, and other electronic devices
Continuous range of values	Discontinuous values
Records sound waves as they are	Converts into a binary waveform
Only used in analog devices	Suited for digital electronics like computers, mobiles and more

PERIODIC SIGNALS

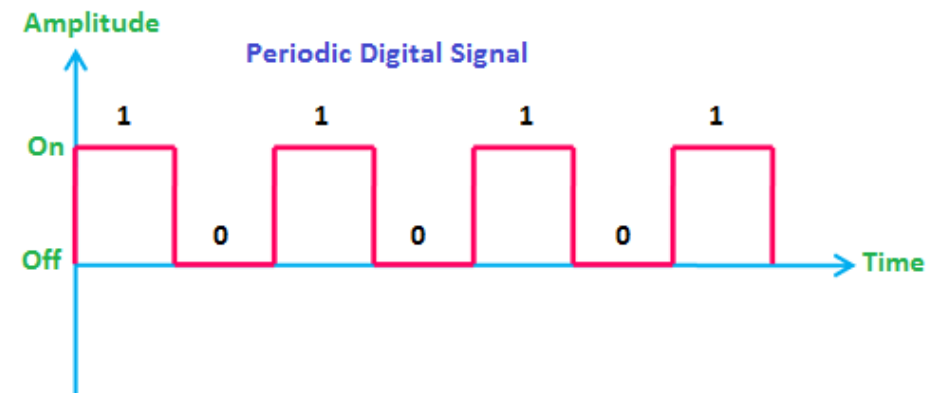
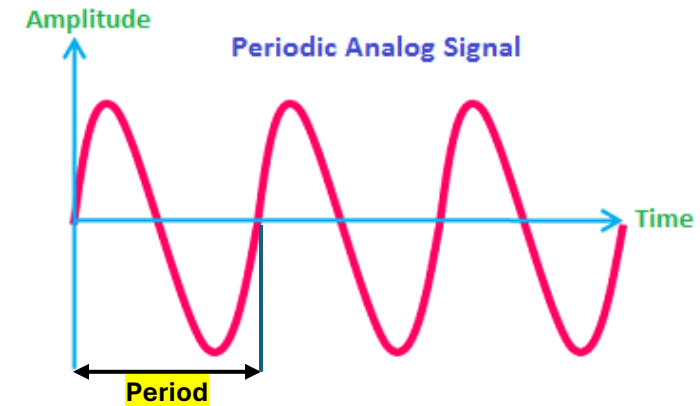
- A periodic signal is a signal that repeats itself after a fixed interval of time, known as the period (T).
- The completion of one full pattern is called a cycle.

Characteristics

- Repeats continuously in a regular pattern.
- Has a fixed period (T).
- Frequency remains constant.
- Commonly represented by sine waves.

Examples

- Alternating Current (AC) power supply.
- Radio and television carrier waves.
- Clock signals in digital systems.
- **Key Point:** A periodic signal satisfies the condition:
$$f(t) = f(t + T)$$
where T is the period of the signal.



APERIODIC SIGNALS

An aperiodic signal is a signal that does not repeat itself over time.

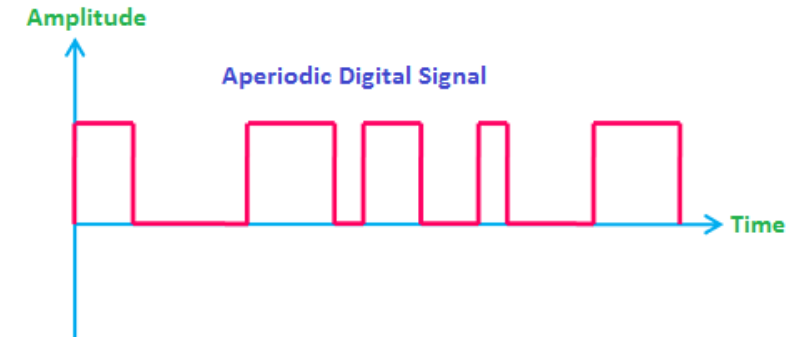
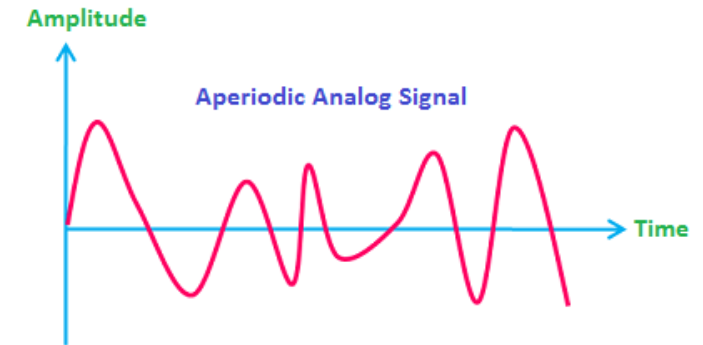
- **Characteristics**

- Irregular waveform.
- No fixed period.
- Frequency varies continuously.
- Often contains multiple frequency components.

- **Examples**

- Human speech.
- Keyboard input.
- Network data traffic.

- **Key Point:** Most information-carrying signals in communication systems are aperiodic in nature.





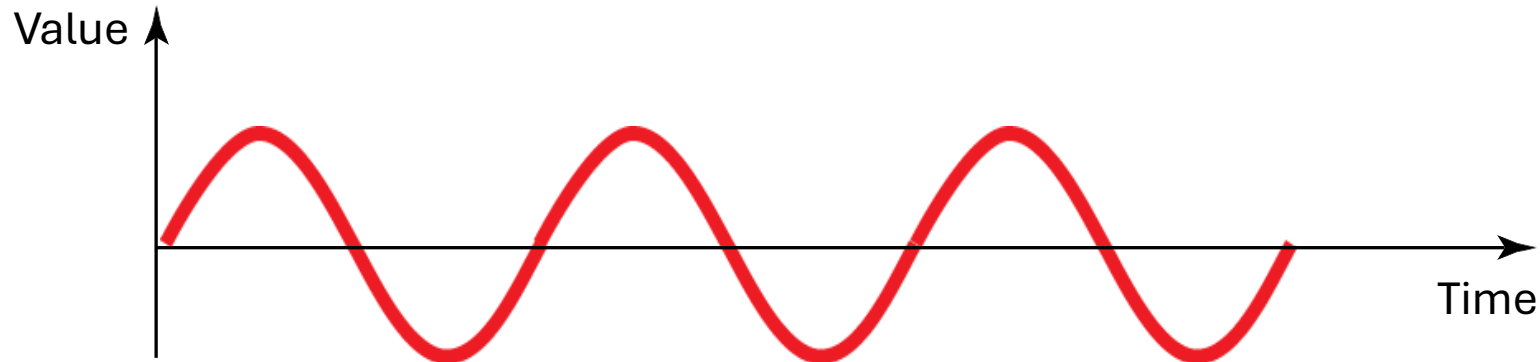
Periodic & Non-Periodic Signals:

- Signals which **repeat itself** after a fixed time period are called Periodic Signals.
- Example of periodic signals is the sine wave.
- Signals which **do not repeat itself** after a fixed time period are called Non-Periodic Signals.
- Example: speech is a non-periodic signal.
- **In data communications, we commonly use periodic analog signals and non-periodic digital signals.**

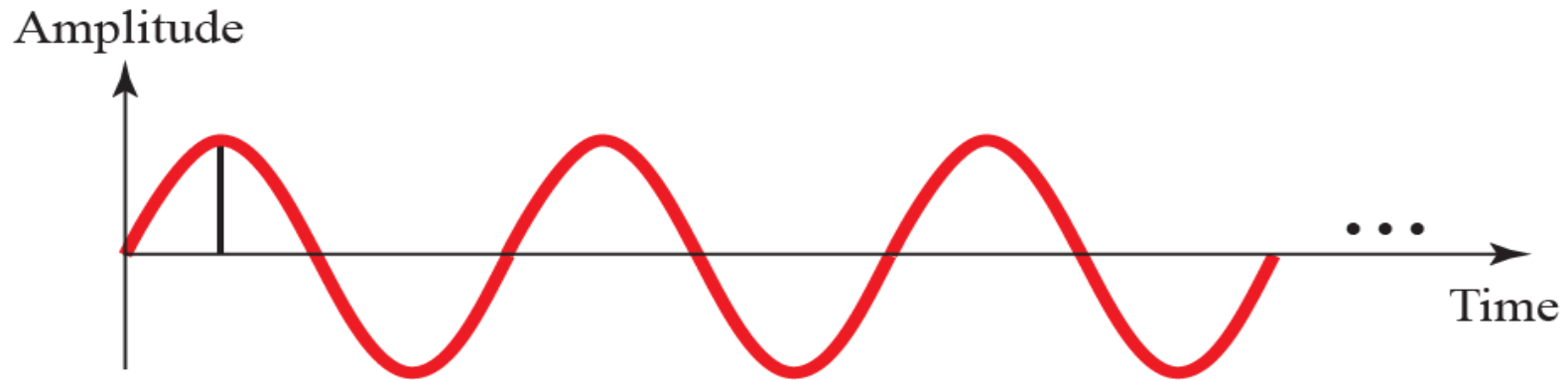


Sine Wave

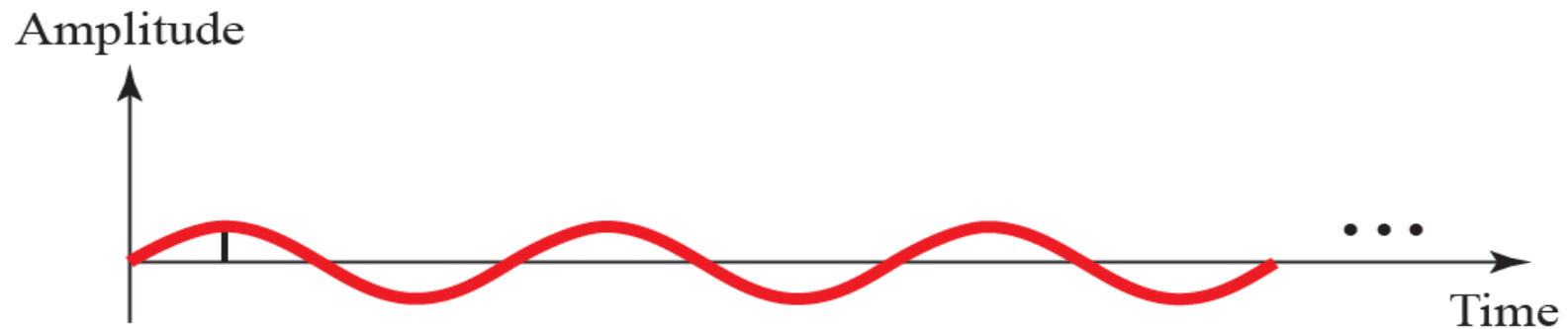
- The sine wave is the most fundamental form of a periodic analog signal.
- When we visualize it as a simple oscillating curve, its change over the course of a cycle is smooth and consistent, a continuous, rolling flow.
- Below shows a sine wave.
- Each cycle consists of a single arc above the time axis followed by a single arc below it.



TWO DIFFERENT SIGNAL WITH DIFFERENT AMPLITUDE



a. A signal with high peak amplitude

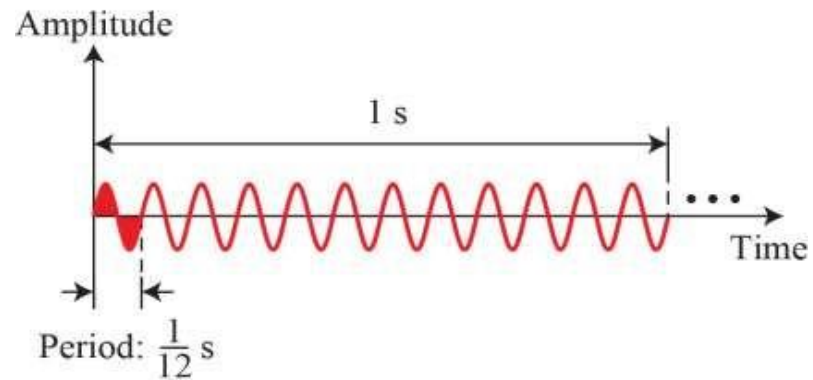


b. A signal with low peak amplitude



TWO SIGNALS WITH THE SAME PHASE AND FREQUENCY, BUT DIFFERENT AMPLITUDES

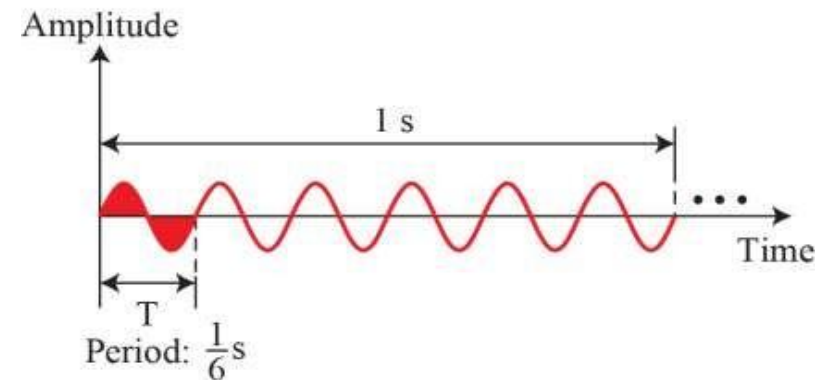
12 periods in 1 s $\rightarrow$ Frequency is 12 Hz



a. A signal with a frequency of 12 Hz

Frequency: Number of Oscillations per second. Measured in Hertz (Hz).

6 periods in 1 s $\rightarrow$ Frequency is 6 Hz



b. A signal with a frequency of 6 Hz

Example

The power we use at home has a frequency of 60 Hz (50 Hz in Europe). The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

This means that the period of the power for our lights at home is 0.0166 s, or 16.6 ms. Our eyes are not sensitive enough to distinguish these rapid changes in amplitude.



Example

The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution: First, we change 100 ms to seconds, and then we calculate the frequency from the period (1 Hz = 10⁻³ kHz).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$
$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

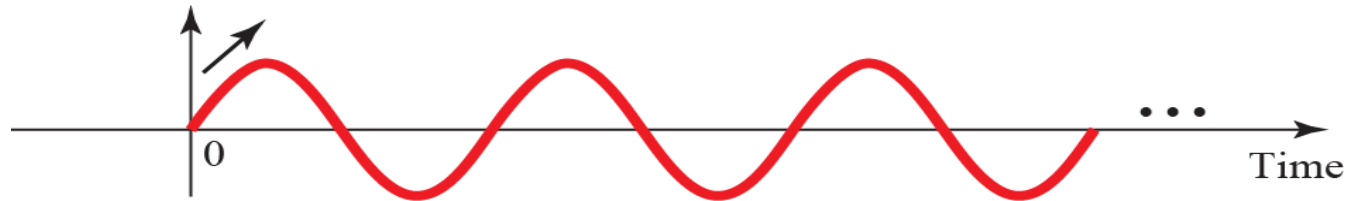


PHASE

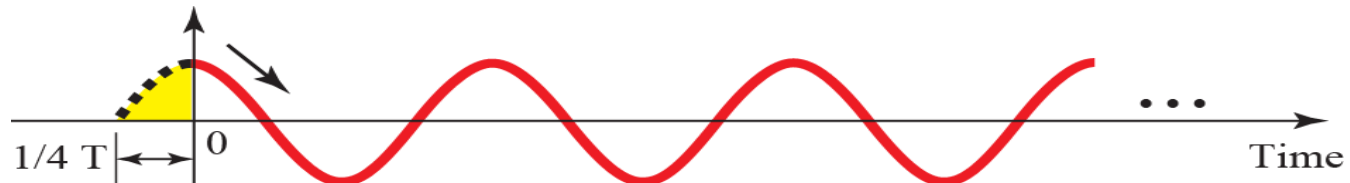
- The term phase, or phase shift, describes the position of the waveform relative to time 0.
- If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift.
- It indicates the status of the first cycle.



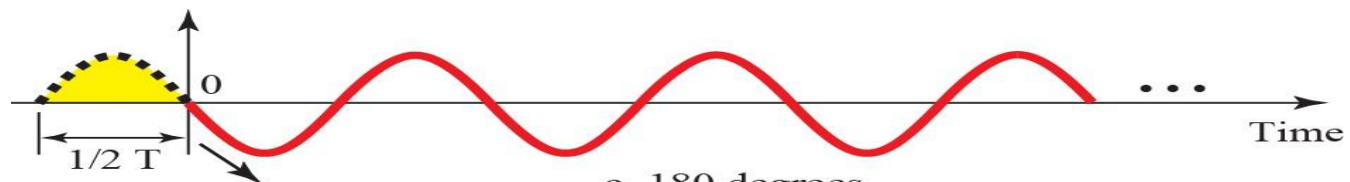
THREE SINE WAVES WITH DIFFERENT PHASES



a. 0 degrees



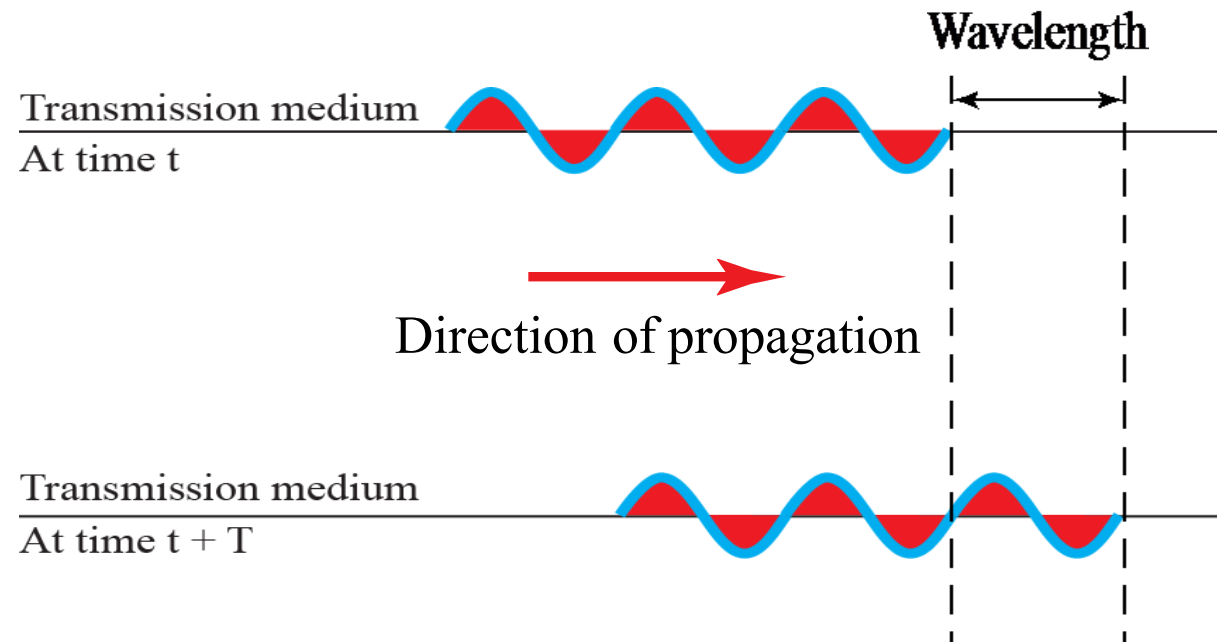
b. 90 degrees



c. 180 degrees

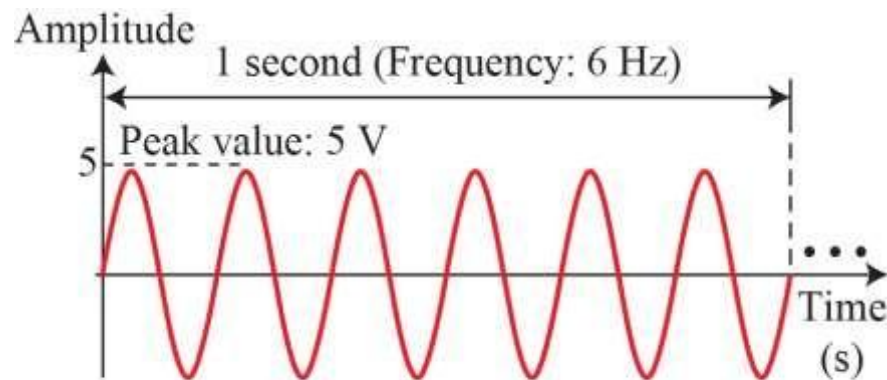
WAVELENGTH

- Wavelength is another characteristic of a signal traveling through a transmission medium.
- Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium.

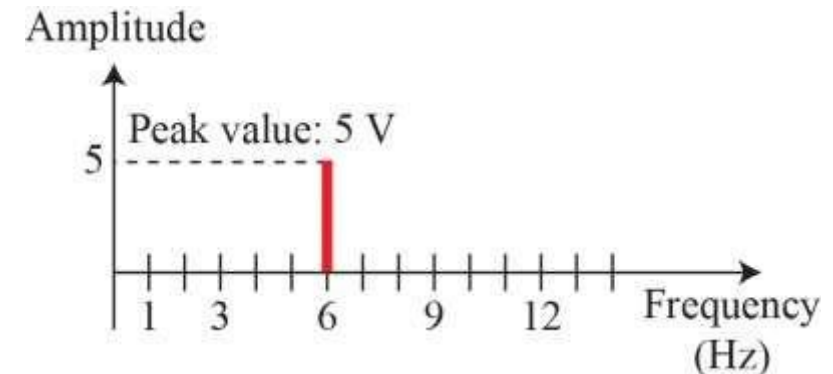


FREQUENCY AND TIME DOMAIN

- A sine wave is comprehensively defined by its amplitude, frequency, and phase.
- We have been showing a sine wave by using what is called a time domain plot.
- The time-domain plot shows changes in signal amplitude with respect to time (it is an amplitude-versus-time plot).
- Phase is not explicitly shown on a time-domain plot.



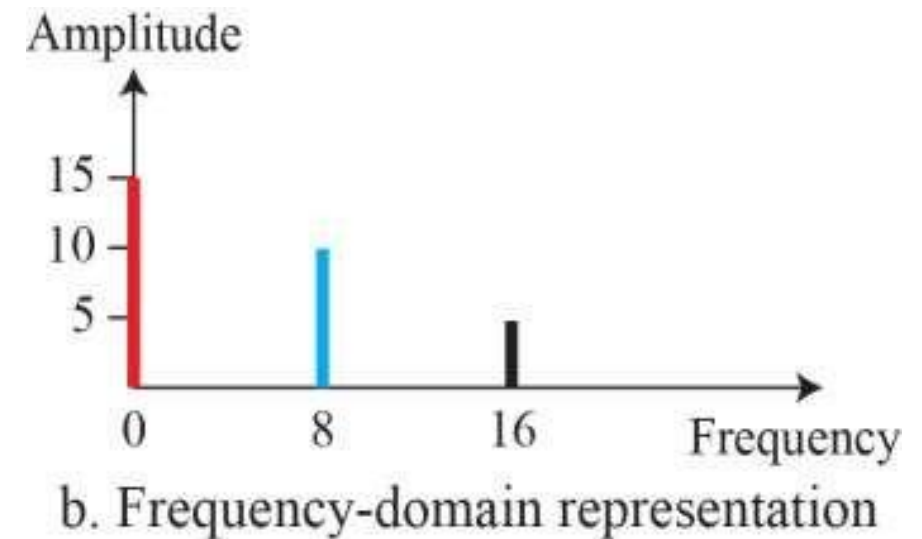
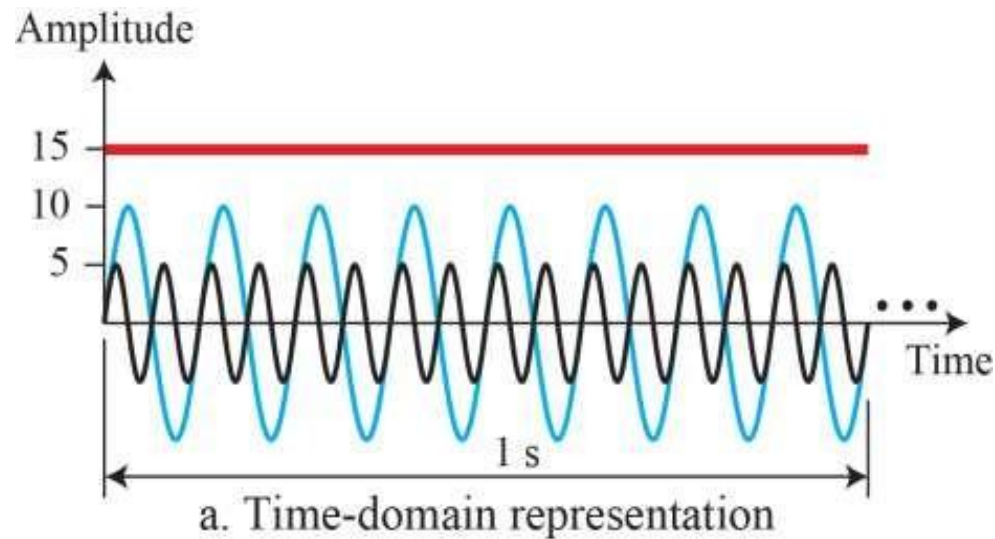
a. A sine wave in the time domain



b. The same sine wave in the frequency domain

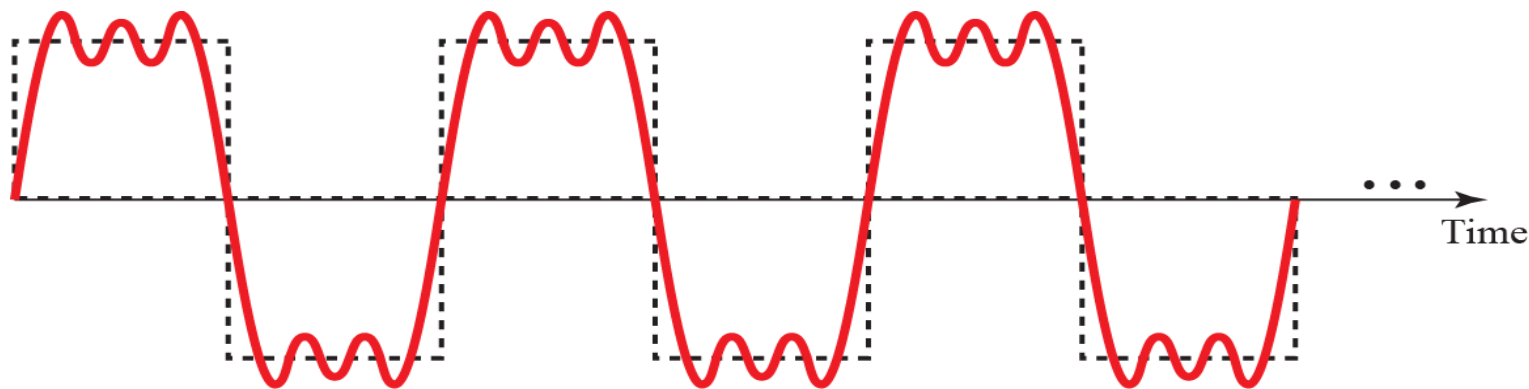
FREQUENCY AND TIME DOMAIN

- The frequency domain is more compact and useful when we are dealing with more than one sine wave.
- For example, the below figure shows three sine waves, each with different amplitude and frequency.
- All can be represented by three spikes in the frequency domain.



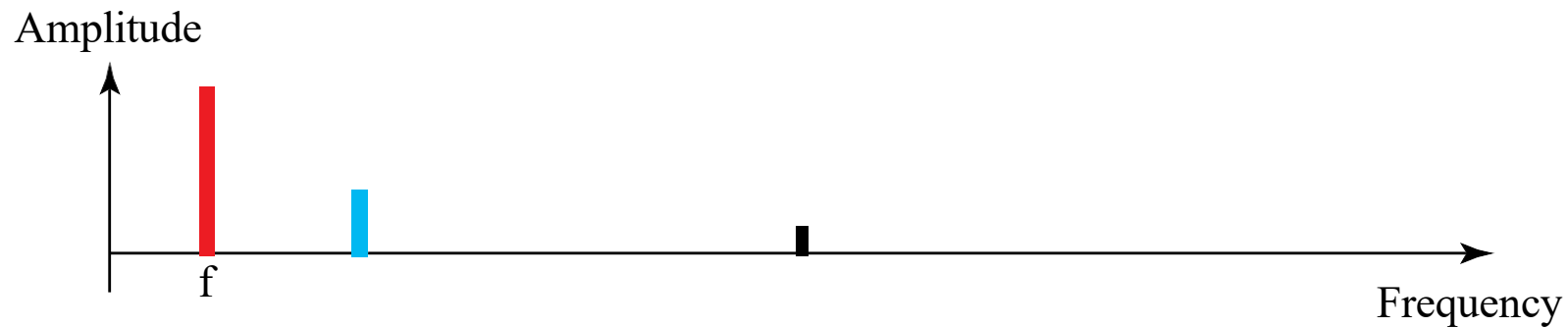
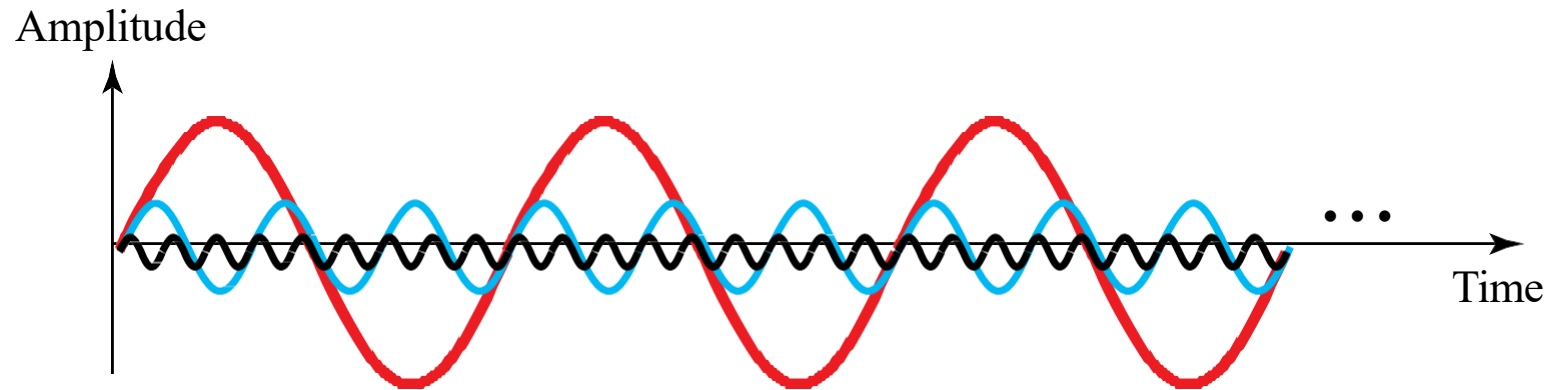
COMPOSIT SIGNAL

- A simple sine wave can be used to carry energy or a single piece of information.
- For example, electric power companies use a 50/60 Hz sine wave to distribute electrical energy.
- Similarly, a single sine wave can be used as an alarm signal in security systems.
- Most signals used in data communication are **composite signals**, formed by combining multiple sine waves with different frequencies, amplitudes, and phases.
- A composite signal can be decomposed into its constituent sine waves using mathematical tools such as Fourier analysis.



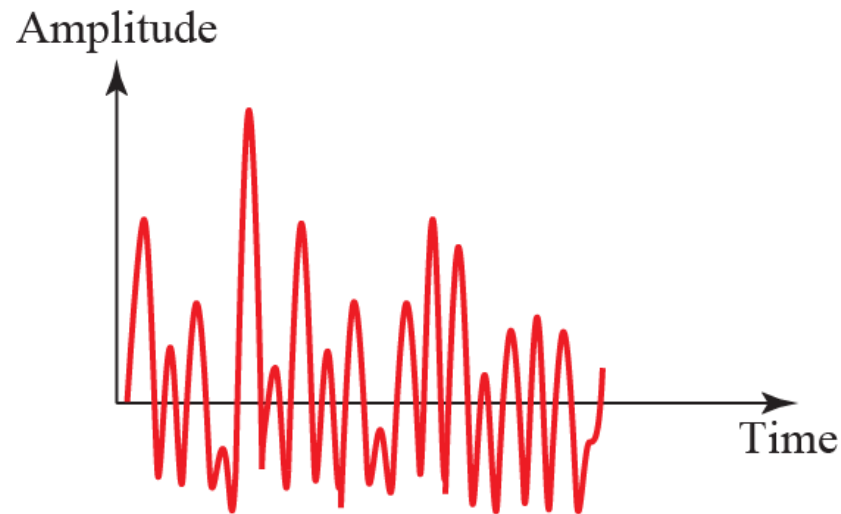


DECOMPOSITION OF A COMPOSITE PERIODIC SIGNAL

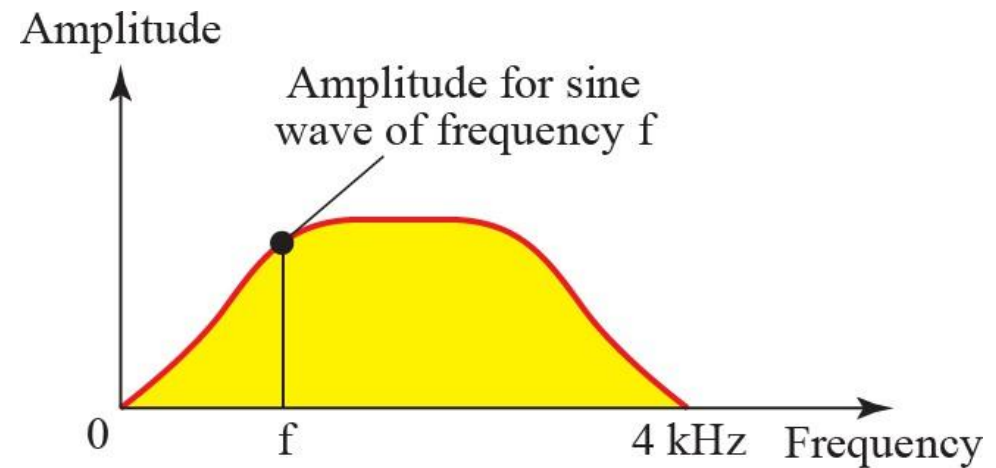


TIME AND FREQUENCY DOMAIN OF A NON-PERIODIC SIGNAL

- The below figure shows a non-periodic composite signal.
- It can be the signal created by a microphone or a telephone set when a word or two is pronounced.
- In this case, the composite signal cannot be periodic, because it implies that we are repeating the same word or words with exactly the same tone.



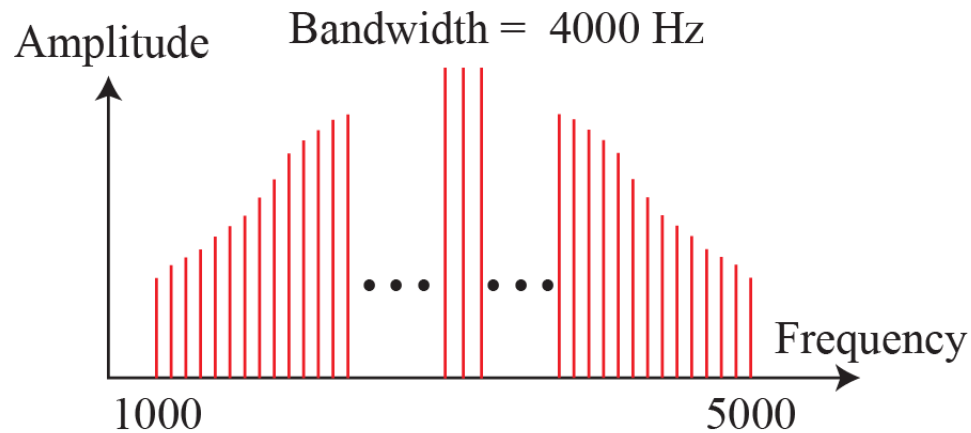
a. Time domain



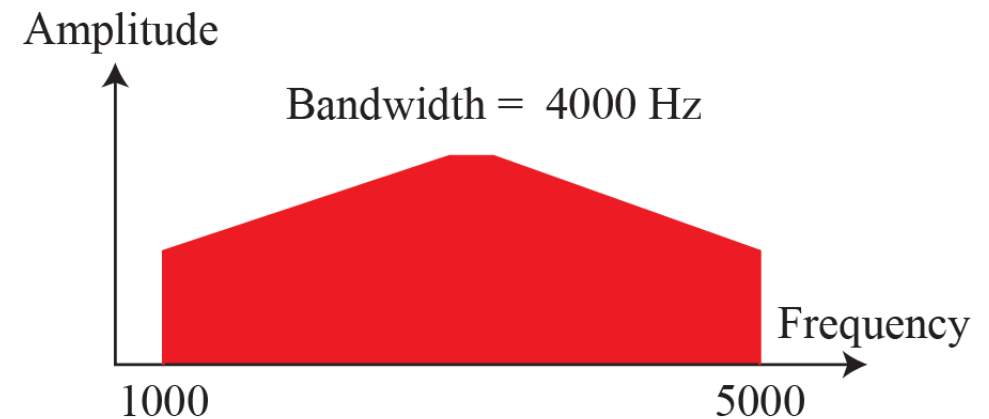
b. Frequency domain

TIME AND FREQUENCY DOMAIN OF A NON-PERIODIC SIGNAL

- The range of frequencies contained in a composite signal is its bandwidth.
- The bandwidth is normally a difference between two numbers.
- For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is $5000 - 1000$, or 4000.



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

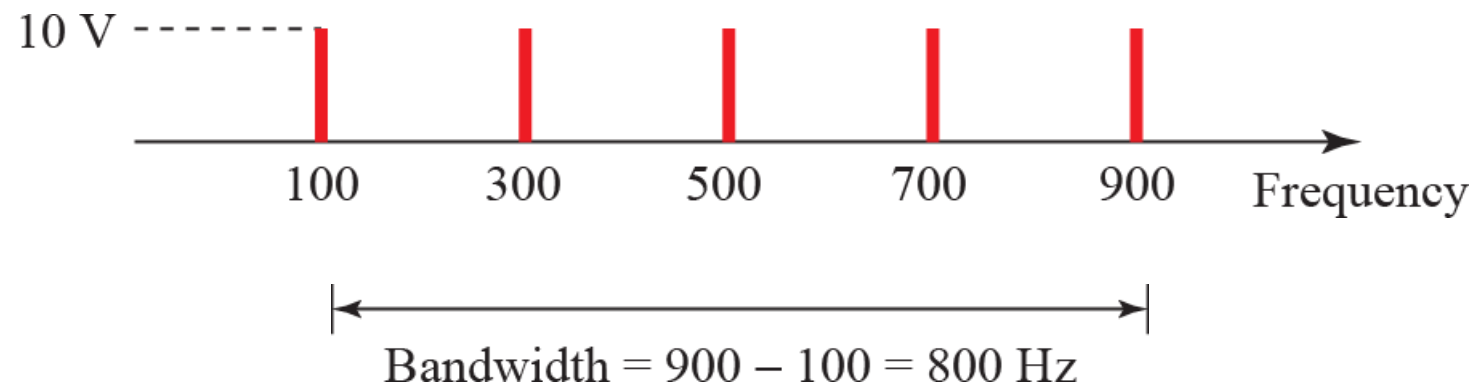


Example

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

Solution: Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$



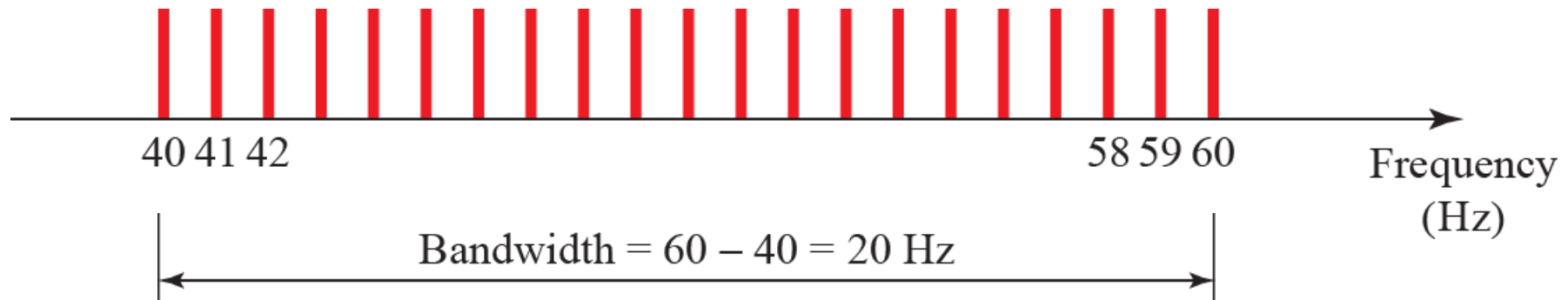
Example

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

Solution: Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \longrightarrow 20 = 60 - f_l \longrightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes.

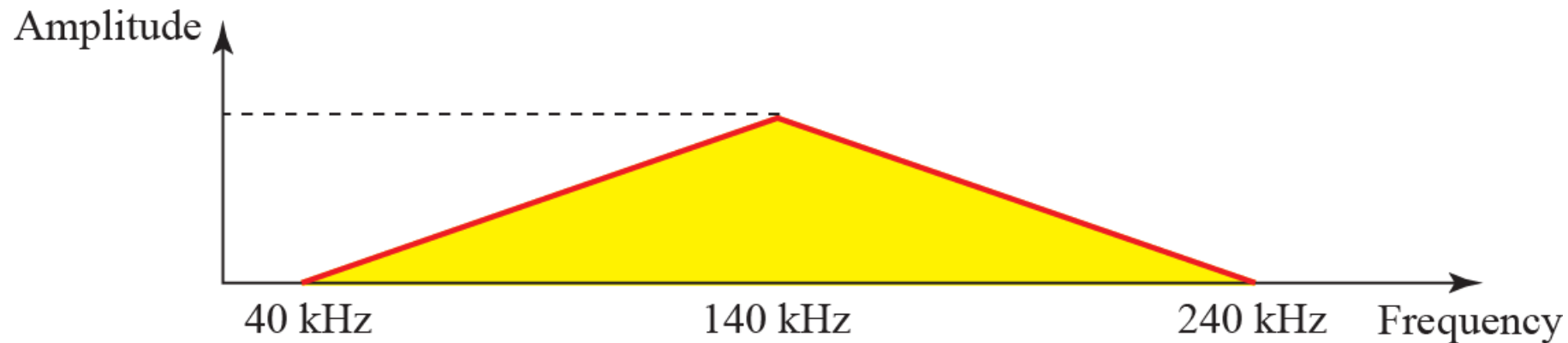




Example

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution: The lowest frequency must be at 40 kHz and the highest at 240 kHz. Below shows the frequency domain and the bandwidth.





DIGITAL SIGNAL LEVELS

- In a digital signal, information is represented using discrete voltage levels.
- Typically, a binary **1** is represented by a positive voltage, while a binary **0** is represented by zero or a lower voltage level.
- The simplest digital signal uses **two levels**, allowing the transmission of one bit per signal element.
- A digital signal can also have **more than two levels**. In such cases, each signal element can represent multiple bits.
- For example, a **four-level digital signal** can represent two bits (00, 01, 10, 11) using four distinct voltage levels.

Key Point: Increasing the number of signal levels allows more bits to be transmitted per signal element, improving data transmission efficiency.

DIGITAL SIGNAL LEVELS

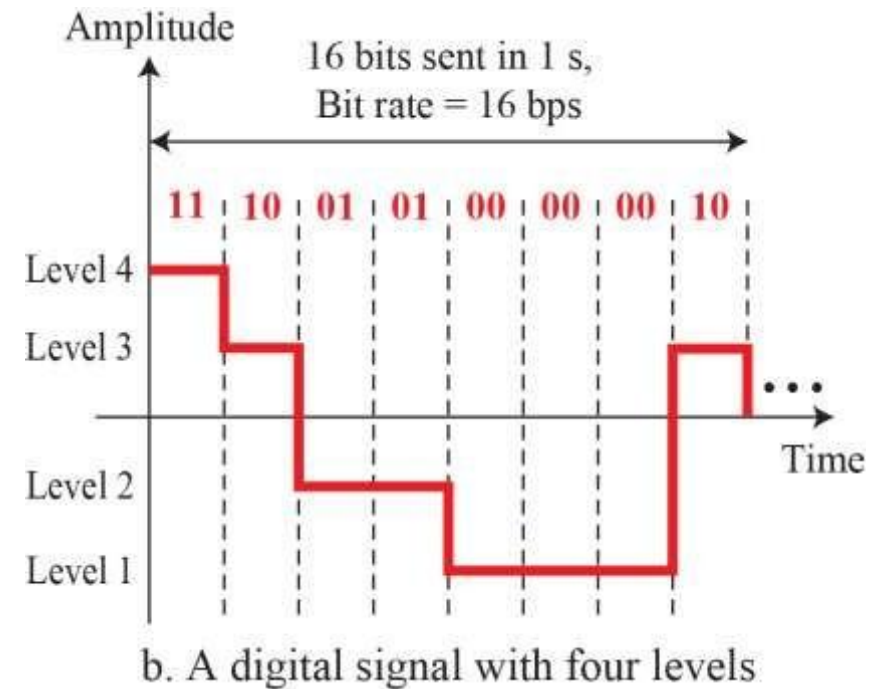
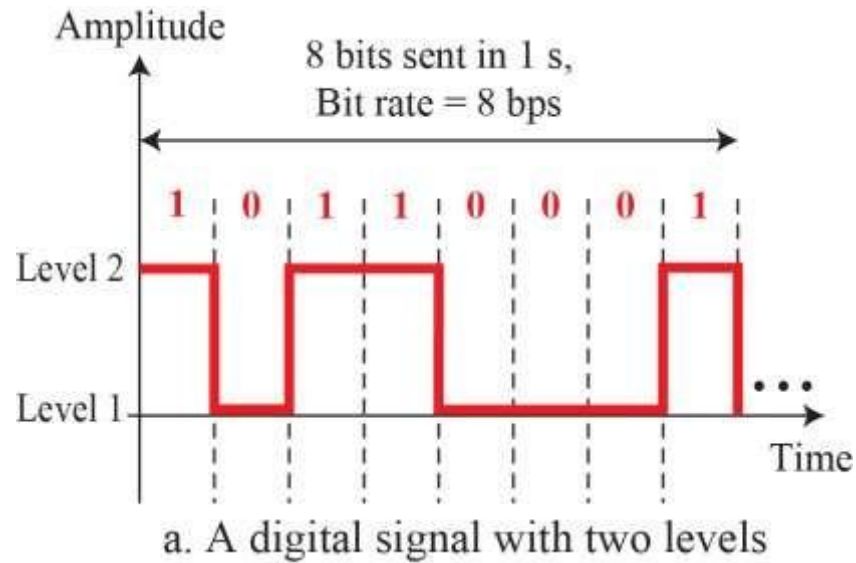


Figure 1.1: Two digital signals: one with two signal levels and the other with four signal levels



Example

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the following formula. Each signal level is represented by 3 bits.

$$\text{Number of bits per level} = \log_2 8 = 3$$

BIT RATE

- Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics.
- Another term—bit rate (instead of frequency)—is used to describe digital signals.
- The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).
- Figure 1.1 shows the bit rate for two signals.
- For an analog signal, **wavelength** is defined as the distance occupied by one complete cycle of the signal on the transmission medium.
- Similarly, for a digital signal, **bit length** is the distance occupied by one bit on the transmission medium during transmission.

$$\text{Bit Length} = \text{Propagation Speed} \times \text{Bit Duration}$$



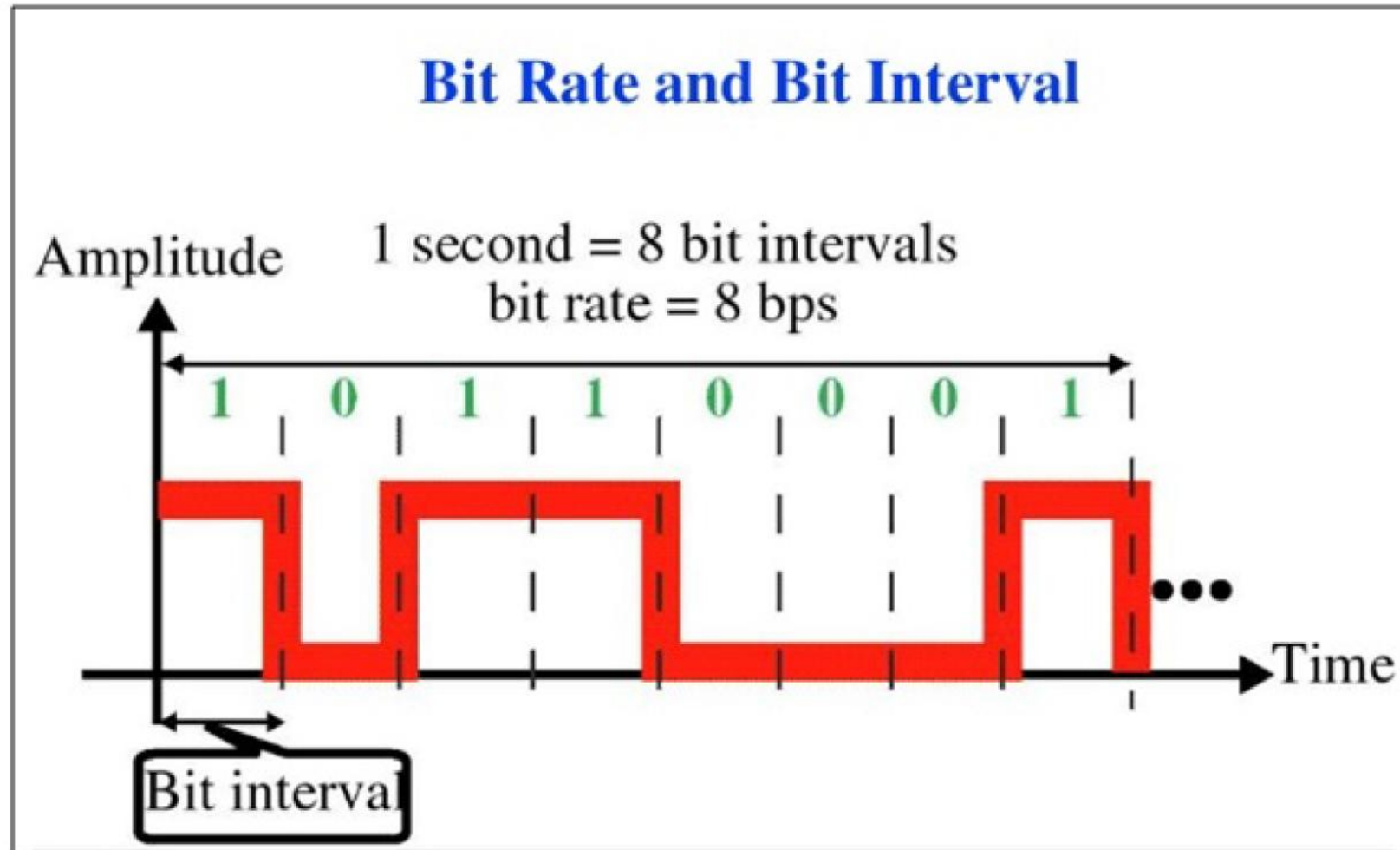
Bit Rate (data rate / Bandwidth):

- It is the number of bits transmitted in one second.
- It is expressed as bits per second (bps).
- Relation between bit rate and bit interval can be as follows
- Bit rate = $1 / \text{Bit interval}$

Baud Rate (signalling speed):

- It is the rate of Signal Speed, i.e., the rate at which the signal / symbol changes.

Bit Rate:



Baud Rate:

- ...is the rate at which signals can change (pulses / second)
- The unit is the baud (Bd), and it is a measurement of speed



- This has a baud rate of 1 Bd, as the signal only changes once a second (taken to be the end)



- This has a baud rate of 3 Bd
- 3 pulses in 1 second



Difference between Bit Rate and Baud Rate

Sr. No.	Key	Bit Rate	Baud Rate
1	Definition	Bit rate is transmission of number of bits per second.	Baud rate is number of signal units per second.
2	Definition	It can be defined as per second travel number of bits.	It can be defined as per second number of changes.
3	Formula	Bit Rate = Baud rate x the number of bit per baud	Baud Rate = Bit rate / the number of bit per baud



Example:

An analog signal has a bit rate of 8000 bps and baud rate of 1000 baud. How many data elements are carried by each signal element?

Answer:

$$S = 1000$$

$$N = 8000$$

$$r = N/S = 8000/1000 = 8 \text{ bits / baud}$$



Thank you!

Doubts and Questions?